# How Far Can General-Purpose LLMs Go in Tigrinya? A Lexicon-Grounded Evaluation with Trivial Baselines

**Anonymous ACL submission**

## Abstract

We evaluate eight general-purpose large language models on three lexicon-grounded Tigrinya tasks (part-of-speech tagging, morphosyntactic labelling, and English→Tigrinya translation) using a 5,783-row annotated bilingual lexicon; a fourth task, lexical alignment, is released with the data but is not well enough posed to carry a result. Our central methodological finding is that such evaluations are easy to over-read, and that on these tasks the apparent competence is an artefact of measurement rather than a property of the models. A permissive matching rule puts the best model at 85.9% on part-of-speech tagging and 89.4% on morphosyntactic labelling, but scoring the same saved outputs by the first word the model commits to costs up to 26 points. More decisively, the prompt supplies an English gloss beside the Tigrinya form, and an oracle that sees only that gloss scores 96.5% and 95.9% on the two tasks: *no model comes within 14 points of what the English alone makes available*, because fewer than one headword in twenty is ambiguous given the English. On translation all eight models score at or below chrF 0.81, while NLLB-200-3.3B, which supports Tigrinya, reaches chrF 16.05 on identical items with an identical scorer. We therefore report a negative result: these tasks, as posed, do not measure Tigrinya competence, and published numbers on them should not be read as doing so. We release the lexicon, the scoring code and the saved model outputs, and argue that the procedure applied here (report every matching rule, bound the task from above as well as below with references that use none of the target language, and quantify prompt echo) requires no resources beyond those an evaluation already has, and should be standard in low-resource evaluation.

## 1 Introduction

Tigrinya is a Semitic language of Eritrea and northern Ethiopia with approximately 9–10 million speakers and very little machine-readable linguistic infrastructure. This scarcity makes it a natural test of whether general-purpose LLMs generalise beyond high-resource languages, and a difficult one to measure: without established benchmarks it is easy to produce numbers that look like competence.

This paper makes three contributions.

1. **A lexicon-grounded benchmark** of 5,783 English-Tigrinya rows (4,331 English headwords, 3,767 Tigrinya forms), each carrying a part-of-speech label, derived from a digitised bilingual dictionary (Union of International Democrats, 2015).

2. **An evaluation of eight LLMs** across POS tagging, morphosyntactic labelling and translation, reported under three matching rules and against references that use no Tigrinya at all. A fourth task in the benchmark, lexical alignment, is reported for completeness but not claimed as a result: 38% of its gold labels are positional fallbacks rather than annotation (Section 5.4).

3. **A negative result about the task design.** The choice of matching rule moves reported accuracy by up to 26 points on fixed outputs, and an oracle restricted to the English gloss scores 96.5% and 95.9% on the two tagging tasks, which no model approaches. The tasks are therefore substantially solvable without the target language, and the numbers they produce, including ours, are not measurements of Tigrinya competence. We give the procedure that exposes this and argue it should be routine.

This lexicon and these four tasks were introduced by Teklehaymanot et al. (2026), whose retraction has been requested. The results below come from new runs on a rebuilt gold set, not from re-scoring that paper's outputs; Section 7 gives the differences.

## 2 Related work

### 2.1 Tigrinya and Ge'ez-script NLP

Tigrinya remains severely underrepresented in NLP: a recent survey of over 50 studies from 2011 to 2025 finds a field whose progress is driven almost entirely by discrete resource-creation milestones rather than by modelling advances (Gaim and Park, 2026), a pattern echoed in broader surveys of Ethiopian-language NLP (Tonja et al., 2023). The resources that exist are small and task-specific. Öktem et al. (2020) built a domain-specific Tigrinya→English NMT system using transfer learning from other Ge'ez-script languages, reporting a 1.3 BLEU improvement over a neural baseline; Tela et al. (2020) showed that an English XLNet fine-tuned on 10k examples reaches 78.88 F1 on Tigrinya sentiment, outperforming mBERT by 7 points; TiGQA contributed 2,685 expert-annotated question–answer pairs from 537 context paragraphs (Teklehaymanot et al., 2024). Tigrinya also appears as one slice of multilingual benchmarks MasakhaPOS, the largest POS dataset for African languages, covering 20 typologically diverse languages under UD guidelines (Dione et al., 2023), and Belebele, a parallel multiple-choice reading-comprehension set in 122 language variants (Bandarkar et al., 2024). The lexicon and the four tasks used here were introduced by Teklehaymanot et al. (2026), whose retraction has been requested; Section 7 states what this paper takes from that work and how the two differ. The lexicon complements the benchmarks above: it is not running text, and it tests the dictionary-headword level that sits beneath all of them. The structural reason such resources are scarce at all is the one Joshi et al. (2020) documented representation in NLP venues tracks resource availability, calling into question the "language agnostic" status of current systems.

### 2.2 LLM evaluation for low-resource languages

NLLB-200 added Tigrinya (`tir_Ethi`, Ge'ez script, Afro-Asiatic/Semitic) to a 200-language translation model (NLLB Team et al., 2022), evaluated on FLORES, whose professionally translated, multilingually aligned design set the standard for low-resource MT benchmarking (Goyal et al., 2022). Against that backdrop, several studies have established that general-purpose LLMs lag purpose-built MT on exactly the languages we study. Robinson et al. (2023) evaluated ChatGPT on 204 FLORES-200 varieties and found NLLB superior on 169 of 201 languages (84.1%), by an average of 11.9 chrF, with the gap widest for African languages and the Ge'ez script specifically; they further report that five-shot prompting improved on zero-shot by only 0.88 chrF on average across 203 directions, which is consistent with our own few-shot probe moving gemma-7b from chrF 0.16 to 0.66 with exact match unchanged. Hendy et al. (2023) reach the same conclusion for GPT models across eighteen directions: competitive for high-resource, limited for low-resource and Zhu et al. (2023) find GPT-4 beating NLLB in 40.91% of directions overall while still trailing badly on the long tail. Adebara et al. (2025) extend this to a continent-scale African benchmark and attribute the residual disparities to data inequities rather than model scale. Our Section 5.3 result, three NLLB checkpoints at chrF 14.1–16.1 against eight LLMs at $\leq 0.81$ is this finding at lexicon granularity, where the LLM floor is not merely lower but absolute.

A separate strand asks whether LLM evaluations in these settings are conducted carefully enough. LAraBench benchmarked 33 Arabic tasks across 61 datasets and explicitly computed a random baseline "to determine if the LLMs predictions are not merely the result of chance" (Abdelali et al., 2023). We regard that step as mandatory rather than optional, and Section 5 shows why.

### 2.3 Trivial baselines and spurious cues

The methodological core of this paper has a direct precedent in natural language inference. Poliak et al. (2018) showed that a hypothesis-only model (one that never sees the premise, a degenerate solution by construction) significantly outperforms a majority-class baseline on a number of NLI datasets; Gururangan et al. (2018) showed the same artefacts let a simple text classifier label the hypothesis alone correctly in about 67% of SNLI and 53% of MultiNLI, concluding that model success "has been overestimated". McCoy et al. (2019) demonstrated that strong test-set scores can rest on fallible syntactic heuristics that collapse on controlled sets, and Niven and Kao (2019) found BERT's 77% on argument reasoning comprehension to be entirely accounted for by spurious statistical cues. Our English-only tagger is a hypothesis-only baseline in this sense: the prompt format `Phrase: {Tigrinya} ({English})` leaves the part of speech largely recoverable from the gloss, and the baseline reaching 60.38 is the lexicon-task

analogue of the 67% SNLI result.

For POS tagging specifically, Kann et al. (2020) make the closest argument to ours: weakly supervised taggers reported to perform "almost as well as supervised ones" were evaluated on languages unlike truly low-resource ones, and on 15 genuinely low-resource languages the best model gets fewer than half the words right, with a strongest baseline macro-average of 39.11%. They also note that such methods presuppose "high-coverage and almost error-free dictionaries" precisely the assumption our Section 3.3 annotator notes undermine, where the source dictionary contains mislabelled parts of speech and outright mistranslations. At the level of benchmark design, Reuel et al.'s assessment of 46 best practices across 24 benchmarks identifies stating random performance as a scored criterion, since without it one cannot separate capability from chance or metric design (Reuel et al., 2024).

### 2.4 Scoring rules, answer extraction and prompt artefacts

Our finding that the matching rule moves reported accuracy by up to 26 points on fixed outputs belongs to a growing literature on evaluation mechanics. Tam et al. (2024) show that constraining LLMs to structured output formats significantly degrades measured reasoning, and disentangle format errors from content by using an LLM as a "perfect parser" rather than a regex the mirror image of our problem, where permissive parsing manufactures competence instead of destroying it. Kirouane and Petrocheilos (2026) document the same failure mode concretely in a low-resource setting: a scorer that scanned the whole response instead of the requested answer line produced a $+29.8$ pp artefact, one-directional against verbose outputs, and they conclude that "any benchmark that requests an answer format must score that format first and report how often it was absent". Our first-token rule is that recommendation applied; our exact / first-token / set-intersection triple reports the spread it would otherwise hide.

The prompt-echo column in Section 5.2 relates to known label-copying behaviour. Zhao et al. (2021) show that few-shot LLM accuracy is destabilised by bias towards answers appearing in the prompt or common in pretraining, varying "from near chance to near state-of-the-art" with prompt format alone. When the gold distribution is 53.7% `noun` and the prompt example contains the word `noun`, a model that copies the example is scored correct by construction under set-intersection gemma-7b does this on 97.4% of answers. We therefore treat echo rate as a reportable diagnostic rather than a footnote.

### 2.5 Metrics and agreement statistics

We report chrF (Popović, 2015) because character $n$-gram F-score is tokenisation-independent and therefore appropriate for Ge'ez-script targets averaging 1.37 whitespace tokens, where word-level BLEU (Papineni et al., 2002) is near-undefined; we follow Post's argument that BLEU is a parameterised family whose configurations are not comparable across papers unless the scheme is pinned down (Post, 2018), and report character-level BLEU alongside chrF for this reason. Agreement is reported as Cohen's $\kappa$ (Cohen, 1960). We note the standard caveat that $\kappa$ is hard to interpret under class imbalance, the "kappa paradox", and that absolute thresholds are not meaningful without a contextual reference (Wong et al., 2021); this is why Section 3.3 reports $\kappa$ beside raw observed agreement and beside the filled-label-only variant, rather than reporting a single coefficient.

### 2.6 Dictionaries as evaluation resources

Digitised bilingual dictionaries are a standard bottom-up resource for languages lacking parallel corpora: Wickramasinghe and de Silva (2023) argue that for low-resource pairs it is "more feasible to move in the bottom-up direction where finer granular pairs such as dictionary datasets are developed first", and Alnajjar et al.'s Ve'rdd addresses the complementary problem of grassroots paper dictionaries edited by many hands, which require re-evaluation rather than direct ingestion (Alnajjar et al., 2020). Our source exhibits exactly the defects that motivate that work: a direction-swapped second block, inline gender markers embedded in the target field, and annotator-flagged mistranslations, and Sections 3.13.3 document them rather than silently cleaning them.

## 3 Dataset

### 3.1 Source and construction

The lexicon was digitised from the Swansea Tigrinya–English dictionary (Union of International Democrats, 2015), with additional entries contributed by native-speaker linguists, and carries the source dictionary's own part-of-speech abbreviations. The digitised source holds 7,234 rows concatenated from two directions, English→Tigrinya

followed by Tigrinya→English, under a single pair of column headers; in the second block the column named `English` therefore holds Ge'ez script. We detect that block by script, reorient it, and merge both directions, keeping distinct senses of one form as separate items *estimate* → ግምት is retained as both noun and verb, which yields **5,783 gold rows** over 4,331 English headwords and 3,767 Tigrinya forms. Three evaluation set sizes follow from this and are not interchangeable: 5,783 gold rows, **5,775 items** for translation and lexical alignment after each task drops duplicate pairs and rows with an empty side, and **5,701 items** for the two tagging tasks, which were run against a further reduction representing each multi-sense entry by its first sense. Appendix A gives the row-level derivation, the provenance of the source file, and the reconstruction of the 5,701-item set from the released gold.

### 3.2 What the annotations do and do not contain

The POS labels are the source dictionary's, not independently re-annotated by us. They are coarse: `noun` accounts for 3,109 of 5,783 labelled entries (53.7%), `verb` 1,542, `adjective` 879.

Morphosyntactic features are **sparse**. Only 185 entries carry gender, number or plurality; the remaining 5,598 (96.8%) carry a value identical to the POS label. We therefore do not treat morphosyntactic labelling as a distinct linguistic dimension, and report it in Section 5.2 as what it measures: a second POS task with a different prompt.

### 3.3 Inter-annotator agreement

A 500-item subset of the lexicon was annotated independently by the first and second authors for part of speech, gender, number, category agreement and lexical alignment. Both are native speakers of Tigrinya. Each assigned labels separately using the annotation guide, and the two sheets were then compared to locate disagreements. The final sheet was compiled from those two sheets: **no third annotator took part, and no party outside the author group adjudicated**. We state this because it bounds what the coefficients below can mean, and because the compiled sheet is sometimes read as an independent adjudication, which it is not.

The records are released as `POS_english_to_tigrigna_Annotated.xlsx` the two annotation sheets, the compiled sheet and the annotation guide that defines each value and `scripts/compute_iaa.py` recomputes every figure below from them.

| Dimension | $\kappa$ | Observed |
|---|---|---|
| Part of speech | 0.9076 | 0.9500 |
| Gender | 0.8079 | 0.9240 |
| Number | 0.6762 | 0.9280 |
| Category agreement | 0.5912 | 0.9000 |
| Lexical alignment | 0.8487 | 0.9480 |

Table 1: Inter-annotator agreement on the 500-item subset, blanks counted as disagreements. All five dimensions are computed over all 500 items.

What the released workbook does not carry is the history of how it was filled. The annotators exchanged their sheets as messaging attachments, and those attachments have since expired, so the files that would carry an edit history are no longer retrievable. The labels themselves are intact and are what the workbook contains; what is missing is the order and timing in which they were entered. We note this because the guide was applied unevenly in ways the figures below expose, and a record of the process would have shown whether that reflects a change of convention partway through.

The five dimensions are not of one kind, and the guide says so. Its instruction is to "verify and annotate" the dictionary entries: gender, number, category agreement and lexical alignment are assigned from scratch against the definitions the guide gives, while the part-of-speech label arrives prefilled from the source dictionary and is verified. The guide treats it as given material elsewhere too, defining category agreement as whether the Tigrinya entry matches "the English item and its POS label".

The two annotators used that latitude very differently. Annotator 1 corrected the dictionary's part of speech on 25 of 500 items; Annotator 2 corrected it on none, letting the source label stand everywhere. All 25 part-of-speech disagreements are therefore of one shape: one annotator exercising the verification the guide asks for, and the other accepting the dictionary.

An annotator sometimes left a cell empty where the other assigned a label. Those are not matching judgements, so we count them as disagreements and report Cohen's $\kappa$ over all 500 items, with a blank treated as its own category (Table 1).

Restricting instead to the items both annotators labelled raises every dimension — 0.9076, 1.0000, 1.0000, 0.9815 and 0.9850 on 500, 420, 428, 445 and 474 items. We report the stricter figures be-

cause the difference between the two columns is largely a difference in what each annotator left blank: Annotator 1 omits 32 gender, 32 number and 46 agreement labels that Annotator 2 supplies, against 6, 4 and 3 in the other direction. Discarding those rows would hide the main source of divergence.

Three properties of this subset bound what the coefficients support.

**The compiled sheet is not an independent third judgement.** This follows from how it was produced, and the records bear it out. The two annotators differ on 28 cells in total: 25 part of speech, one category agreement and two lexical alignment. On the 25 part-of-speech items the compiled sheet reproduces Annotator 2 on all 25 and Annotator 1 on none; on the other three it reproduces Annotator 1. **On none of the 28 does it record a label that neither annotator assigned.** Since Annotator 2's part-of-speech column is the source dictionary's throughout, the compilation rejected all 25 of Annotator 1's proposed corrections and deferred to the dictionary in every case. It selects between the two existing columns rather than adding a judgement of its own, so agreement with it is not evidence about either annotator, and it cannot serve as an adjudicated gold standard.

A third annotator would not repair this. The coefficients in Table 1 are computed between the two annotation columns and would be unchanged by a third. The divergence between those columns is in any case not a matter of conflicting labels: on gender and number the annotators never differ on an item both labelled, and what separates them is how often each left a cell empty. What the subset would need is a fresh annotation, stratified over the categories and including verbs, by annotators outside the author group; we report that as future work in Section 9 rather than adjudicating the present records after the fact.

**Agreement on filled labels is near-total.** Where both annotators assigned a gender or number value, they never differ 0 disagreements across 420 and 428 items. The divergence between them is entirely in coverage, not in choice of label, which is why $\kappa$ falls as far as it does once blanks count.

**The subset is not a random sample.** Tracing each item to its source sheet shows the 500 are the alphabetical head of each category: all 9 prepositions, all 10 pronouns and all 6 interjections, then 305 adjectives and 150 nouns from the tops of those sheets. The corpus is 54% noun and 25% verb; this subset is 61% adjective and **is drawn from no verb entries at all**. The only part-of-speech cells reading `Verb` anywhere in the records are seven of Annotator 1's corrections to noun headwords. The coefficients therefore describe agreement on this subset and are not an estimate for the corpus. Establishing one requires a stratified re-annotation covering verbs, which we leave to future work (Section 9).

The workbook also carries the annotators' free-text notes on 201 of the 500 items, documenting mislabelled parts of speech in the source dictionary, inflected forms given as citation forms, and outright mistranslations (*at least* → ᑭᑭᐱᓗᒃ, which means *at most*). These support the caveat in Section 3.2 that the POS labels are the source dictionary's and were not independently verified.

## 4 Experimental setup

Eight models, all loaded through the Hugging Face `transformers` interface (Wolf et al., 2020) and run on a single compute node with four A100 GPUs, 32 CPU cores and 128 GB of RAM (Table 2).

**We do not compare architectures.** All five causal models are instruction-tuned and all three sequence-to-sequence models are raw pretrained checkpoints; the two factors are perfectly confounded in this model set and no result here can separate them. The three base seq2seq models receive span-infilling prompts matched to their pretraining objective; they nonetheless produce almost no usable output, which we report as a property of those checkpoints on this task rather than of their architecture.

Generation is greedy with `max_new_tokens=128`, `min_new_tokens=1`. `xlm-roberta-base` (Conneau et al., 2020) was included initially but fails on all tasks (its prompts contain no `<mask>` token) and is excluded.

### 4.1 Scoring

Model answers are frequently verbose: gemma-7b's median POS answer is 14 words. We report three rules.

- **exact** the cleaned answer equals the gold label.
- **first-token** the model's first word is a gold label word. This is the answer the model commits to, and is our headline rule.

| Label | Checkpoint | Arch. | Inst. |
|---|---|---|---|
| gemma-2b | `google/gemma-2b-it` (Gemma Team, Google DeepMind, 2024) | causal | yes |
| gemma-7b | `google/gemma-7b-it` (Gemma Team, Google DeepMind, 2024) | causal | yes |
| mistral-7b | `Mistral-7B-Instruct-v0.2` (Jiang et al., 2023) | causal | yes |
| qwen-7b | `Qwen1.5-7B-Chat` (Bai et al., 2023) | causal | yes |
| falcon-7b | `falcon-7b-instruct` (Almazrouei et al., 2023) | causal | yes |
| mt5-small | `google/mt5-small` (Xue et al., 2021) | seq2seq | no |
| mt5-large | `google/mt5-large` (Xue et al., 2021) | seq2seq | no |
| byt5 | `google/byt5-small` (Xue et al., 2022) | seq2seq | no |

Table 2: Evaluated models. Architecture and instruction-tuning status are perfectly confounded in this set.

- **set-intersection** any word of the answer is a gold label word. This is permissive: an answer naming several parts of speech is correct as soon as one fits. We report it only to show the gap.

### 4.2 Baselines

The prompt supplies the English gloss alongside the Tigrinya form (`Phrase: {Tigrinya} ({English})`), so a system can score well without using the Tigrinya at all. We therefore bound the two tagging tasks from below and from above with references that see no Tigrinya whatsoever.

- **Majority** always answer the most frequent gold label (`noun`).
- **English-only tagger** an NLTK (Loper and Bird, 2002) English POS tagger applied to the gloss alone. This is what one off-the-shelf English system achieves, and is a lower bound on what the gloss makes available.
- **English-only ceiling** an oracle that sees only the English string and answers with the most frequent gold label that headword carries anywhere in the evaluation set. This is the best any English-only system can reach, and we report it because the lower bound alone understates how much of the task the gloss already determines.

The ceiling is high: 96.49 on part-of-speech tagging and 95.86 on morphosyntactic labelling. Only 189 of 5,025 headwords (3.8%) carry more than one gold part-of-speech label, and 212 (4.2%) more than one morphosyntactic label, so the Tigrinya form can supply information the gloss does not already give on fewer than one item in twenty. A model scoring below the ceiling has not shown that it used the Tigrinya.

Differences against the tagger baseline are tested with a one-sided binomial test at $n = 5{,}701$;

| Model | exact | **first-tok** | set-int |
|---|---|---|---|
| mistral-7b | 33.71 | **81.56** | 85.91 |
| gemma-7b | 0.00 | **79.39** | 82.44 |
| gemma-2b | 71.71 | **72.86** | 73.29 |
| falcon-7b | 26.94 | **40.64** | 44.45 |
| qwen-7b | 17.00 | **17.00** | 17.07 |
| mt5-large | 0.00 | 0.09 | 1.75 |
| byt5 | 0.00 | 0.00 | 0.05 |
| mt5-small | 0.00 | 0.00 | 0.00 |
| **Majority (`noun`)** | | **53.73** | |
| **English-only tagger** | | **60.38** | |
| **English-only ceiling** | | **96.49** | |

Table 3: POS tagging accuracy (%) under three matching rules, against the two English-only references. No model reaches the ceiling.

`scripts/task_ceiling.py` computes the ceiling, the ambiguity counts and the tests.

## 5 Results

Three tasks carry results. The fourth, lexical alignment, is reported in Section 5.4 so that its outputs are on the record, but its gold labels are too unreliable to support a conclusion.

### 5.1 POS tagging ($n = 5{,}701$)

Three models exceed the tagger baseline, each significantly ($p < 10^{-4}$): mistral-7b by 21.19 points, gemma-7b by 19.01 and gemma-2b by 12.49. Falcon-7b and qwen-7b fall below the majority baseline; all three seq2seq checkpoints are at or near zero.

**No model reaches the English-only ceiling.** The best, mistral-7b, falls 14.93 points short of it; gemma-7b 17.10 and gemma-2b 23.63. Since the ceiling is what a system achieves from the gloss alone, the ordering above is not evidence that any model used the Tigrinya form. It is consistent with

| Model | **first-tok** | set-int | echo |
|---|---|---|---|
| gemma-2b | **64.59** | 78.02 | 1.2% |
| gemma-7b | **63.32** | 89.42 | **97.4%** |
| mistral-7b | **54.92** | 81.09 | 3.7% |
| falcon-7b | **39.17** | 76.28 | 19.5% |
| qwen-7b | **34.33** | 54.45 | 0.4% |
| mt5-small | 0.86 | 2.49 | 0.0% |
| mt5-large | 0.11 | 0.74 | 0.0% |
| byt5 | 0.00 | 38.55 | 69.7% |
| **Majority** | **52.38** | | |
| **English-only tagger** | **61.57** | | |
| **English-only ceiling** | **95.86** | | |

Table 4: Morphosyntactic labelling accuracy (%). The final column is how often an answer contains the prompt's own example. No model reaches the ceiling.

| Model | chrF | $\text{BLEU}_c$ | $\text{BLEU}_w$ | exact |
|---|---|---|---|---|
| **NLLB-3.3B** | **16.05** | **14.23** | 0.24 | **16.24** |
| NLLB-1.3B | 15.01 | 13.29 | 0.23 | 15.00 |
| NLLB-600M | 14.11 | 12.59 | 0.25 | 14.42 |
| byt5 | 0.81 | 0.01 | 0.00 | 0.05 |
| gemma-2b | 0.39 | 0.01 | 0.01 | 0.07 |
| mistral-7b | 0.26 | 0.00 | 0.00 | 0.03 |
| mt5-small | 0.20 | 0.04 | 0.02 | 0.05 |
| gemma-7b | 0.16 | 0.00 | 0.00 | 0.03 |
| mt5-large | 0.13 | 0.04 | 0.01 | 0.02 |
| falcon-7b | 0.13 | 0.00 | 0.00 | 0.03 |
| qwen-7b | 0.04 | 0.00 | 0.00 | 0.28 |

Table 5: Translation quality. All values on a 0–100 scale.

every model using the gloss and using it worse than an English-only oracle would.

The exact column shows how brittle these numbers are. Gemma-7b scores 0.00% exact and 79.39% first-token: it always answers in a sentence, never a bare label. A reader given only one of these columns would draw opposite conclusions.

### 5.2 Morphosyntactic labelling ($n = 5{,}701$)

Because the gold is 96.8% identical to the POS label (Section 3.2), this is effectively a second POS task.

Only two models exceed the tagger baseline of 61.57: gemma-2b by 3.02 points ($p < 10^{-4}$) and gemma-7b by 1.75 ($p = 0.003$). Both margins are statistically significant at $n = 5{,}701$, and both are small. The remaining six fall below the baseline, and five below the majority baseline of 52.38.

Against the ceiling the picture is uniform: **the best model is 31.28 points below what the English gloss alone makes available**, and the eight models span 31 to 96 points below it. Whatever separates gemma-2b from mistral-7b here, it is not evidence of morphosyntactic analysis of Tigrinya.

The final column measures how often an answer contains the prompt's own example, `preposition, noun, singular`. Gemma-7b echoes it in 97.4% of answers; because the example contains `noun`, and `noun` is the majority gold label, set-intersection scores those echoes as correct. Its apparent 89.42 is largely an artefact of copying the prompt. ByT5's 38.55 is the same artefact at 69.7% echo, and drops to 0.00 under first-token.

### 5.3 Translation, English→Tigrinya ($n = 5{,}775$)

Tigrinya references average 1.37 whitespace tokens, so word-level BLEU-4 is near-undefined; we report chrF and character-level BLEU (Table 5).

All eight general-purpose models are at the floor. NLLB-200, for which Tigrinya (`tir_Ethi`) is a supported target language, reaches chrF 16.05 on identical items with an identical scorer, roughly twenty times the best LLM.

Note that word-level BLEU-4 is 0.24 for NLLB-3.3B, which translates 16.24% of items exactly. **Word-level BLEU is not usable at lexicon scale**, and a paper reporting it would show a working system as scoring zero.

We verified the LLM failure is not one of output format. Few-shot prompting with eight exemplars and an explicit format constraint moved gemma-7b from chrF 0.16 to 0.66 with exact match unchanged at 0.00; extracting the Ge'ez span from verbose answers recovered the reference in 2 of 5,775 items. The median character-level similarity between model output and reference is 0.000: the models emit Ge'ez-shaped strings that are not the target words (ባህል for ብዙሕ, ልልል as repetition).

### 5.4 Lexical alignment

We report this task for completeness but do not draw conclusions from it: 2,208 of 5,783 alignment labels (38%) are positional fallbacks (`AlignmentSource: auto`) rather than annotation. Format accuracy ranges from 0.0383 (mt5-large) to 0.5484 (gemma-2b); semantic accuracy from 0.0012 (mt5-small) to 0.8876 (mistral-7b). The two measures diverge widely for individual

models, which makes the task ill-posed as currently scored.

## 6 Discussion

**The matching rule is a reporting decision, not an implementation detail.** On fixed model outputs, moving from set-intersection to first-token costs gemma-7b 3.05 points on POS and 26.10 on morphosyntax; moving to exact match costs it everything. A paper that reports one number without naming the rule has not reported a result.

**Bound the task from above, not only from below.** A lower-bound baseline tells you whether a model beats a weak system; it does not tell you whether the task needs the target language. On POS the majority answer scores 53.73 and an English tagger 60.38, and three models clear both, which reads as cross-lingual competence until the upper bound is added. An oracle restricted to the English gloss reaches 96.49, and the best model is 14.93 points below it. The same holds on morphosyntax, where the gap is 31.28. Reporting only the lower bounds would have let this paper claim a result it does not have.

**Prompt echo is a measurable confound.** Gemma-7b reproduces the prompt's example in 97.4% of its morphosyntax answers, and because the example contains the majority gold label, a permissive scorer rewards it. We recommend reporting echo rate wherever a prompt contains a worked example.

**Purpose-built MT still dominates at the lexicon level.** NLLB-200 checkpoints reach chrF 14.11-16.05 where the best general-purpose model reaches 0.81 more than an order of magnitude on chrF. For low-resource languages, model selection dominates model size.

**What we can claim about Tigrinya competence.** On the two tagging tasks, nothing. Three models beat the English tagger on POS and two on morphosyntax, each significantly at $n = 5{,}701$, but all eight sit well below what the English gloss alone affords, and fewer than one headword in twenty is ambiguous given that gloss. These tasks cannot separate a model that reads Tigrinya from one that reads the English and answers worse than an oracle would. Translation is the one task here that cannot be solved from the prompt's English, and there the result is unambiguous: no model translates into Tigrinya. Even NLLB, which supports the language, reaches only 16.24% exact on dictionary headwords, so the difficulty is real rather than an artefact of our setup.

## 7 Relation to prior work on this lexicon

This lexicon and these four tasks were introduced by Teklehaymanot et al. (2026), whose retraction has been requested.

The results reported here are not a re-scoring of that paper's outputs. They come from new runs carried out in September 2026, against a gold set rebuilt from the source lexicon, and the two differ in their inputs as well as in their scoring. The earlier runs evaluated 3,899 items per tagging task; these evaluate 5,701. The earlier gold carried the source dictionary's parenthetical notation, so an expected tag was stored as `(adv) adverb` rather than `adverb`, and an entry was presented with all of its comma-separated Tigrinya senses rather than one. Prompts built from those entries differ, and so do the generations they elicit.

Every figure in this paper is computed from the lexicon, the saved generations and the scoring code released with it and listed in Section 8. No number is carried over from the earlier paper, and the two sets of results are not alternative measurements of the same quantity.

## 8 Reproducibility

The lexicon, the gold label files, the saved model outputs, the annotation workbook and the scoring code accompany this submission as supplementary material, and will be released publicly on acceptance. Every number in this paper is recomputed from those artifacts by the commands in Table 6; none requires re-running a model. `$ROSTER` denotes the eight evaluated models, which the scoring script takes as an explicit argument so that a reported table is reproduced over exactly that set.

The supplementary material also contains the raw generation for every item, stored per task and per model, so any scoring rule other than the three reported here can be applied to the same outputs without re-running the models.

## 9 Limitations

- POS labels are the source dictionary's, verified but not independently re-annotated. The agreement coefficients in Section 3.3 describe a 500-item subset that is alphabetical and drawn from

| Reported in | Command (in `scripts/`) | Artifact read |
|---|---|---|
| §3.1 counts | `build_merged_eval_set.py --dry-run --verify` | `Combined_POS_Lexicon.csv` |
| §3.1 5,701 set | `build_sense_split_eval_set.py --verify` | `*_sense_split.json` |
| §3.3 agreement | `compute_iaa.py` | `*_Annotated.xlsx` |
| §4.2 ceiling | `task_ceiling.py` | `*_sense_split.json` |
| §5.1, §5.2 | `rescore_tasks.py --models $ROSTER` | `rescored_scores.json` |
| §5.3 LLMs | `compute_bleu.py` | `translation_fidelity_bleu.json` |
| §5.3 NLLB | `probe_nllb_translation.py` | `nllb_probe_*.json` |
| §5.3 few-shot | `probe_fewshot_translation.py` | `fewshot_probe_*.json` |

Table 6: Each reported table, the command that regenerates it from the released artifacts, and the artifact it reads. No step requires model inference; all are computed from the saved generations.

no verb entries, annotated by the first and second authors rather than by annotators external to the work, and compiled without a third adjudicator. They are not an estimate for the corpus, and a stratified re-annotation covering verbs, by annotators outside the author group, would be needed for that.

- No edit history survives for the annotation. The annotators exchanged their sheets as messaging attachments, and those attachments have since expired, so the files that would carry a version history are no longer retrievable. The labels are intact in the released workbook; what cannot be evidenced is the order and timing in which they were entered.
- Morphosyntactic features cover 185 of 5,783 gold rows.
- Lexical alignment gold is 38% positional fallback; Section 5.4 is not a result.
- Entries are dictionary headwords, not running text; performance here does not predict sentence-level performance.
- The model set cannot separate architecture from instruction-tuning. An instruction-tuned seq2seq model (`flan-t5`, `mt0`) would be needed.
- Both tagging tasks are substantially solvable without Tigrinya: an English-only oracle reaches 96.49 and 95.86, and under 5% of headwords are ambiguous given the gloss. This is a property of the task design, not of the models, and it bounds what any result on these tasks can mean. A task that withheld the gloss, or that drew its items from the ambiguous subset, would measure something the English does not already determine; we have not run that experiment.
- The ceiling is computed from the evaluation set itself, so it is an oracle rather than a system: it knows the label distribution of each headword. It bounds what the English makes available on these items, not what an English-only system would achieve on unseen data.

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## A Dataset derivation

This appendix records the row-level derivation summarised in Section 3.1. Every count is reproducible from the released code: `build_merged_eval_set.py --dry-run --verify` rebuilds the gold files from the source CSV and compares them against what is released, and `build_sense_split_eval_set.py --verify` does the same for the 5,701-item set.

**Source and provenance.** The digitised file `Combined_POS_Lexicon.csv` holds 7,234 rows: 3,587 English→Tigrinya entries followed by 3,647 Tigrinya→English entries. It is a flattening of the working annotation workbook, whose eleven part-of-speech sheets hold the same 7,234 rows with the same per-category counts (3,929 noun, 1,784 verb, 993 adjective, 194 adverb, and so on). The flattening lifts the inline gender and number markers that the workbook keeps inside the Tigrinya field `(nf)` ወካሲት, ከሳሲት out into the label column, which is the origin of the finer categories `(nm) noun masculine`, `(nf) noun feminine` and `(npl) noun plural`; the two files differ on exactly the 247 rows carrying such a marker. The direction swap is present in the workbook as well, in 3,635 of its 7,234 rows (50.2%), so it originates in digitisation rather than in the export.

**7,234 counts rows, not independent annotations.** Of the 3,646 unique reverse-direction pairs, 1,448 (39.7%) also occur in the forward block.

**From 7,234 to 5,783.** Merging the two directions deduplicates on the (English, Tigrinya, features) triple rather than on the pair alone, so that distinct senses of one form survive: *estimate* → ግምት is kept as both noun and verb, and *grocer* → በዓል ድኳን as both `noun` and `noun, masculine`. Expanding comma-separated senses gives 5,864 atomic pairs.

**From 5,783 to 5,775.** Each task script applies its own pair-level deduplication when loading the gold, dropping 7 duplicate (English, Tigrinya) pairs and 1 row with no English side.

**From 5,775 to 5,701.** The two tagging tasks were run against a gold file in which each multi-sense entry is represented by its first sense: (*a little*, ንእሽቶይ, ቁሩብ, ውሕድ) was evaluated as (*a little*, ንእሽቶይ). That intermediate file was not originally released, so the task scripts as published would evaluate 5,775 items rather than the 5,701 behind Sections 5.1 and 5.2. We reconstruct it from the released gold and verify the result row-for-row against the saved model outputs: 5,701 items, no row missing, none extra, and no label mismatch. The reconstructed file is released alongside the gold.